SS1 PHYSICS LESSON NOTES (THIRD TERM)
WEEK 5:

DEFINITION OF WAVE

A wave is a disturbance that travels through a medium or space, carrying energy without transferring matter.

Waves originate from vibrations and are found everywhere in nature.

Examples include light, sound, and water waves.

DIFFERENT PARTS OF A WAVE

1. Rest Position

This is the position of particles when no wave passes through them. It is the equilibrium line.

2. Displacement

This is the distance a particle moves from its rest position.

3. Amplitude

The maximum displacement from the rest position.

4. Crest and Trough

- Crest = highest point of the wave
- Trough = lowest point of the wave

5. Wavelength (λ)

The distance between two consecutive points in phase (e.g. crest to crest).

6. Frequency (f)

Number of waves passing a point per second.

Unit: Hertz (Hz)

7. Period (T)

Time taken for one complete wave.

Formula:

$T = 1 / f$

8. Wave Speed (v)

Speed at which a wave travels.

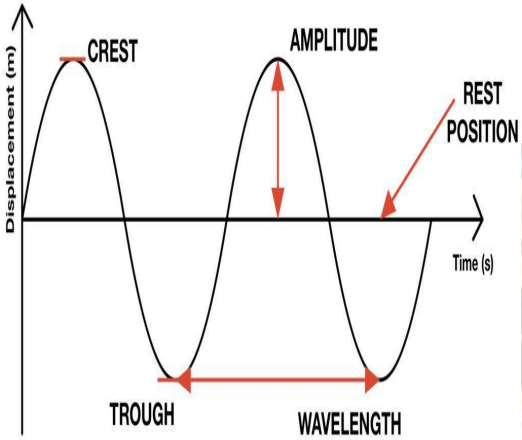
Formula:

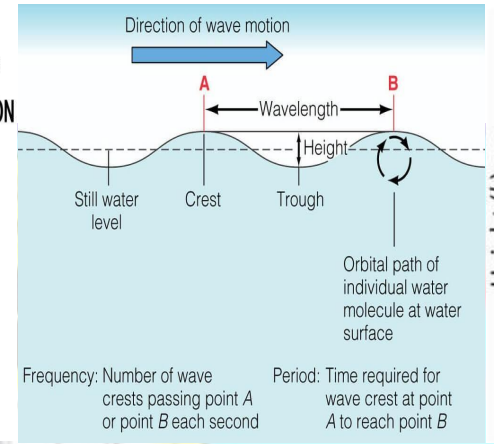
$v = f \times \lambda$

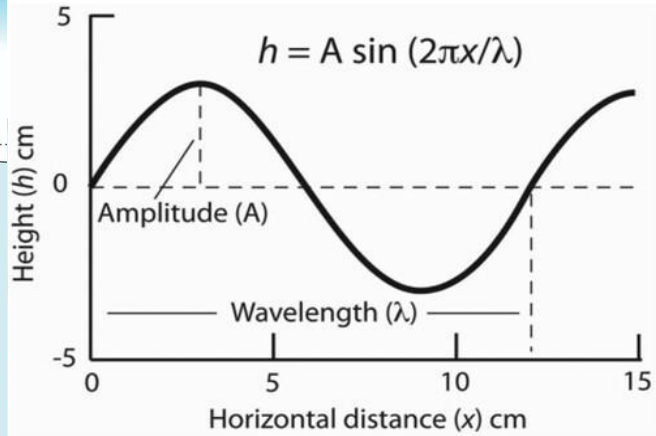
Where:

- v = wave speed (m/s)
- f = frequency (Hz)
- λ = wavelength (m)

VISUAL STRUCTURE OF A WAVE







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CLASSIFICATION OF WAVES

1. Based on Medium

Type	Requires Medium	Examples
Mechanical Waves	Yes	Sound, water waves
Electromagnetic Waves	No	Light, radio waves

Mechanical Waves

Require a medium to travel.

Examples:

- Sound waves
- Water waves
- Rope waves

Electromagnetic Waves

Do not require a medium.

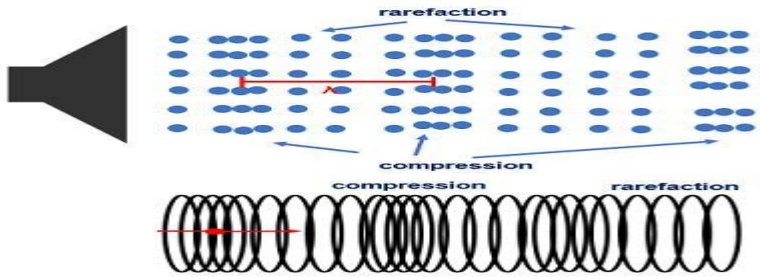
Examples:

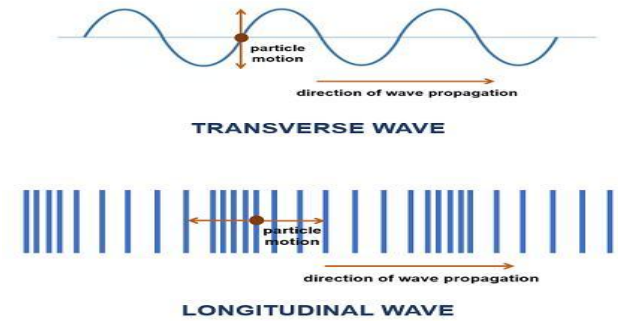
- Light
- X-rays
- Radio waves

TYPES BASED ON PARTICLE MOTION

Type	Particle Motion	Examples
Transverse	Perpendicular	Light, water waves
Longitudinal	Parallel	Sound waves

VISUAL COMPARISON OF WAVES





DETAILED TYPES OF WAVES

1. TRANSVERSE WAVES

Particles move perpendicular to wave direction.

Features:

- Has crests and troughs
- Travels in solids and liquid surfaces

Examples:

- Light waves
- Water waves
- Rope vibrations

2. LONGITUDINAL WAVES

Particles move parallel to wave direction.

Features:

- Has compressions and rarefactions
- Travels through solids, liquids, and gases

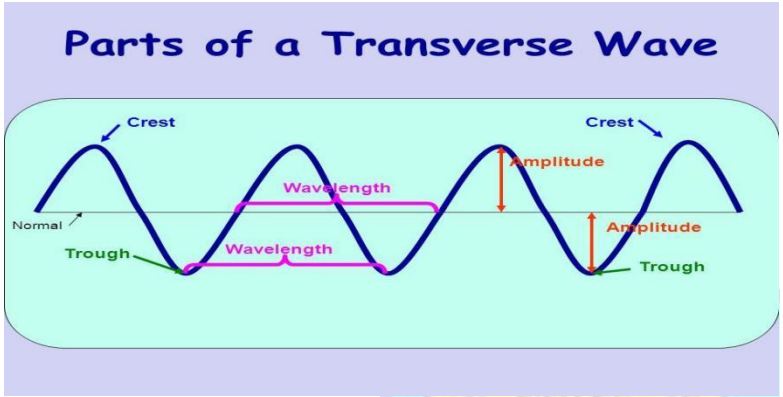
Examples:

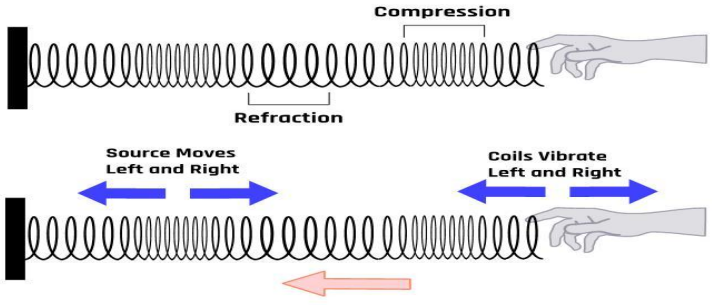
- Sound waves
- Slinky waves

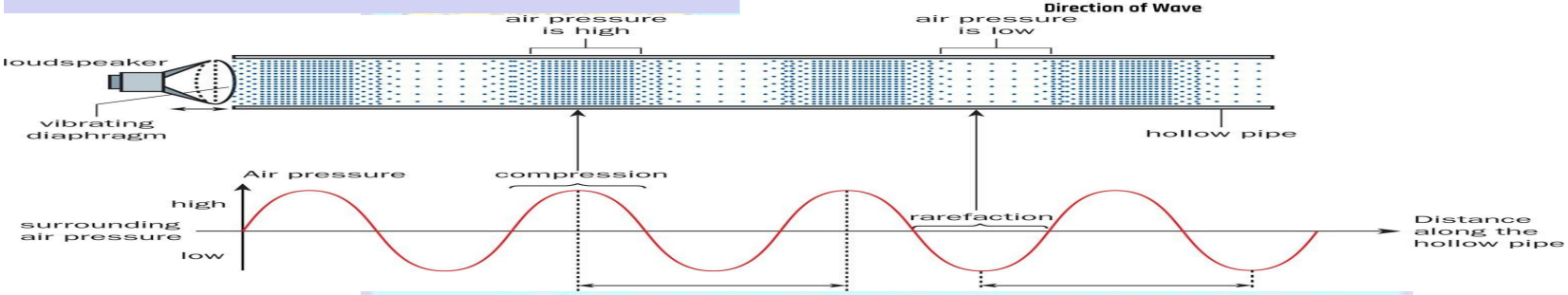
DIFFERENCES BETWEEN WAVES

Feature	Transverse	Longitudinal
Particle motion	Perpendicular	Parallel
Shape	Crest & trough	Compression & rarefaction
Medium	Solids & surface liquids	All states
Example	Light	Sound

WAVE DIAGRAMS (FULL VISUAL)







GOD IS OUR STRENGTH

EVALUATION

1. Define a wave
2. Differentiate mechanical and electromagnetic waves
3. Draw transverse and longitudinal waves
4. Give examples of each wave type
5. Calculate wave speed:

$v = f \times \lambda$
$v = 5 \times 2 = 10$ m/s

ASSIGNMENT

1. Draw:
    - One transverse wave
    - One longitudinal wave
2. Label:
    - Crest, trough
    - Compression, rarefaction
3. Research 5 applications of electromagnetic waves